EXECUTIVE SUMMARY: Business Intelligence and Sales Performance Analysis -Superstore Dataset

Project Overview

This project involved analysing the Superstore sales dataset using Microsoft Power BI to demonstrate how Business Intelligence can support business performance monitoring and decision-making.

The dataset contained 9,994 transaction records across 21 columns, covering the period 2014–2017. The analysis focused on sales, profitability, customer segments, product categories, geographical performance and monthly sales trends.

The data was prepared using Power Query, with data types reviewed and quality checks performed for missing values, duplicate records, blank rows, sales values, discount values and date consistency. An interactive Power BI dashboard was then developed to present the results through key performance indicators and visual analysis.

Key Performance Results

Key Performance Indicator	Result
Total Sales	$2,297,200.86
Total Profit	$286,397.02
Total Orders	5,009
Average Order Value	$458.61
Profit Margin	12.47%

Key Business Findings

The analysis identified several important patterns:

1. **Technology was the strongest category**, generating $836,154.03 in sales and $145,454.95 in profit, with a profit margin of 17.40%.

2. **Furniture showed a profitability concern.** Although it generated $741,999.80 in sales, its profit was only $18,451.27, resulting in a margin of 2.49%.

3. **Office Supplies demonstrated comparatively strong profitability**, recording $719,047.03 in sales and a profit margin of 17.04%, despite having the lowest sales among the three categories.

4. **Product profitability varied considerably.** Copiers recorded a profit margin of approximately 37.19%, while Machines achieved approximately 1.79%. Tables, Bookcases and Supplies also recorded negative profit margins.

5. **Customer and monthly sales patterns varied.** Consumer customers contributed approximately 50.56% of total sales. Monthly performance fluctuated, with periods of growth followed by declines and recoveries rather than a consistent upward trend.

Recommendations

Based on these findings, the following actions are recommended:

- Review Furniture pricing, discounting and supplier costs to improve profitability.
- Investigate loss-making sub-categories, particularly Tables, Bookcases and Supplies.
- Explore opportunities to grow sales of higher-margin products such as Copiers and Accessories.
- Develop targeted strategies to increase Corporate and Home Office customer sales while retaining Consumer customers.
- Use monthly sales patterns to improve demand forecasting, inventory planning and resource allocation.

Conclusion

The project demonstrates how Business Intelligence and interactive dashboards can transform raw sales data into useful information for management decision-making. While the Superstore dataset indicates positive overall profitability, the results also highlight differences in performance across product categories, sub-categories, customer segments and time periods.

The most important management priority is to focus not only on increasing sales, but also on improving the profitability of products and categories that generate weaker returns. Further investigation into pricing, discounts, costs and customer demand would help establish the underlying causes of these performance differences.

The Power BI dashboard provides a foundation for ongoing performance monitoring and more evidence-based business decisions.